TRAN NHAT NAM

Product Data Engineer | Ho Chi Minh City, Vietnam

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Summary

Product Data Engineer with 4+ years of experience creating **end-to-end data solutions** – from **ETL/ELT pipelines** and **data platforms** to **data governance, BI and AI applications**. More than a technician: I dive deep into the business needs of each department (Sales, Marketing, Finance, BOD) to turn data into decisions – **cutting ETL operational cost by 90%**, **consultation time by 60%**, and **accelerating decisions up to 3x**.

Technical Skills

- **Languages:** Python, SQL
- **Data Engineering:** dbt (medallion modeling, tests, docs), Dagster, Airflow, dlt, Airbyte, n8n, Apache Kafka, ETL/ELT
- **Databases & Warehouses:** PostgreSQL, BigQuery, DuckDB, StarRocks, dimensional modeling (star schema), data lineage, vector databases
- **BI & AI/ML:** Looker, Metabase, LLM APIs, RAG & vector search, prompt engineering, HuggingFace, ASR (Wav2Vec 2.0)
- **Cloud, DevOps & Deployment:** AWS Lambda, Google Kubernetes Engine (GKE), Helm, Docker, GitHub Actions, CI/CD, Linux, Vercel, Railway, Dokploy, Coolify
- **Integration & Business Systems:** REST APIs, web scraping (Scrapy, BeautifulSoup), Odoo, ERPNext, Haravan, BizFly, AppSheet
- **Stakeholder Communication:** translating business requirements into technical specs, executive presentations, self-serve dashboards

Experience

Product Data Engineer | Jemmia Diamond **Sep 2023 – Present**

- **[Data Platform & Orchestration]** Architected and deployed a scalable Modern Data Stack from scratch using Dagster (deployed via Dokploy) as software-defined assets. Engineered automated ingestion pipelines with dlt, Airbyte, and n8n to centralize data from 12+ fragmented sources (Social Ads, Haravan, ERP, Stringee) into a PostgreSQL Data Warehouse – 150 raw tables processing 500K+ rows/day.
- **[Data Modeling & Transformation]** Designed scalable Star Schema (Kimball) models and authored 200+ transformation models across layered stages (Raw → Staging → Intermediate → Marts). Reduced pipeline development and maintenance time by 90% while ensuring 100% data accuracy through dbt tests (schema, not-null, accepted-values, relationships, freshness) and enforcing PII masking and lineage via OpenMetadata.
- **[Serverless ETL & IoT Integration]** Migrated brittle legacy workflows (Zapier) to event-driven architectures on AWS Lambda (deployed via AWS SAM). Designed a high-throughput pipeline integrating hardware RFID scanners with e-commerce systems via webhooks (~1,000 records / 3MB per scan, 20 scans/day), enabling time-series analysis for inventory optimization and supply-chain management.
- **[Enterprise Systems & Master Data]** Led the migration and implementation of enterprise systems: migrated 10GB+ legacy data to ERPNext and engineered a custom Product Information Management (PIM) system on NocoDB (40+ tables) integrated with a Cloudflare R2 data lake. Executed zero-downtime backfill for 8,000+ SKUs and set up CDC – reducing cross-departmental manual operations by 40% with 97% user adoption.
- **[GenAI & Data Activation]** Transformed BI by migrating the analytics layer from Looker to Metabase and building two AI products on the warehouse: **Salesaya** (AI sales assistant cutting consultation time 60%) and **Metabot** – a Telegram/Discord agent for secure natural-language data queries without exposing the underlying database (Metabase API, Next.js, React, Redis context caching).
- **[DataOps & Leadership]** Established CI/CD pipelines for all analytical applications and ERP deployments, and configured real-time anomaly alerting via Sentry. Led data-literacy workshops for the BOD and department heads, driving a data-driven culture and engineering standards across the organization.

Data Analyst (Contract) | BabyBet (Startup) **Jul 2023 – Sep 2023**

- **Contracted to assess a mobility venture's feasibility:** consolidated market & competitor data on 5+ players (Grab, be) from 15+ market-intelligence, survey and secondary-research sources into structured datasets.
- **Delivered 10+ demand, willingness-to-pay, pricing and unit-economics analyses** that informed the cross-functional go/no-go decision and phased strategy.

Data Analyst | AVLI Group JSC **Feb 2023 – Jul 2023**

- **Built the ingestion layer:** AWS Lambda scraping workers (Python, BeautifulSoup, REST APIs) for batdongsan.com and 7+ other sites, plus the company's CSV/Excel/Sheets file sources via **Airbyte** – loading 300K+ property records into PostgreSQL.
- **Designed a 15-table dimensional data warehouse** (star schema) as a single source of truth for customers, properties and interactions, orchestrated end-to-end with **Airflow**.
- **Built a low-code CRM on AppSheet** (customer tracking, attendance, call logging for customers/landlords, apartment images, care history) – **cutting customer-care and property-lookup time by 50%**.
- **Delivered 10+ Looker dashboards** for customer demand, inventory and sales performance, used by 25+ users.

Data Research Assistant | AISIA Research Lab **Jun 2022 – Jan 2023**

- Labeled 80+ hours of Vietnamese speech data and helped optimize **ASR models (Wav2Vec 2.0)**, cutting word error rate by ~**12%** to narrow the gap with FPT's solution.

Personal Projects

FB Feed Crawler (Multi-Provider Scraping Platform) **2026**
Technologies: Python, FastAPI, PostgreSQL, APScheduler, Docker, Playwright/Crawl4AI repo

- Built a resilient Facebook Page/Group crawler with a 3-engine fallback chain (Crawl4AI GraphQL browser → Scrapling → facebook-scraper) behind circuit breakers, staying stable against Facebook's constant UI/API changes.
- Scheduled daily incremental runs with APScheduler, DB-backed crash recovery, cookie-pool rotation with health scoring, and a FastAPI surface to manage targets, monitor provider health, and export data.

CryptoStream Labs (Automated ELT Pipeline) **2024**
Technologies: Google Cloud (GKE, BigQuery), Apache Airflow, dbt, Airbyte, Docker, Helm, SQL repo

- Built an automated ELT pipeline ingesting cryptocurrency data from the CoinMarketCap API via Airbyte into BigQuery, then transformed raw data into a partitioned One Big Table (OBT) with dbt to speed up reporting.
- Orchestrated the end-to-end workflow on Apache Airflow for hourly freshness, deploying the full stack on Google Kubernetes Engine (GKE) via Helm for high availability.

Sign2Text (Traffic-Sign Recognition & Vietnamese OCR) **2022**
Technologies: Python, YOLOv7, VietOCR, OpenCV, Streamlit repo

- Led a university scientific-research team building a traffic-sign recognition system: labeled the dataset and trained a YOLOv7 detector, then extracted Vietnamese text from detected signs with VietOCR and played it back as audio for drivers.
- Shipped an end-to-end Streamlit demo covering detection → OCR → voice output, iterating the model across multiple training experiments.

Hackathon & Open-Source Side Projects **2024–2026**
Technologies: Next.js, TypeScript, Docker, Airflow, Airbyte, pg_duckdb, StarRocks repos

- **Hackathon:** built SAIGONXUA, an anti-marketing jewelry app that analyzes each product against a shopper's budget and needs and explains why to buy or avoid it (Next.js, Prisma/SQLite, Gemini via OpenRouter).
- **Open source:** published ready-to-deploy Docker templates running heavyweight data tools on lightweight PaaS/VPS – Airflow, Airbyte, pg_duckdb (PostgreSQL + DuckDB), StarRocks.

Education

University of Information Technology, VNU-HCM **2019 – Jun 2023**
Bachelor of Data Science Ho Chi Minh City, VN
GPA: 7.9/10 | TOEIC 725